1. A bank wants Claude to assist relationship managers. Data residency requires EU processing only, and PII must not leave approved systems. Which architectural constraint is most critical to bake into the design?

A. Select approved regional endpoints/deployment options, minimize PII in prompts, and keep sensitive systems behind controlled tools with policy enforcement
B. Use the cheapest global endpoint regardless of region
C. Store all prompts indefinitely in an unencrypted shared drive
D. Paste full CRM records into every prompt for maximum context

Correct:
Select approved regional endpoints/deployment options, minimize PII in prompts, and keep sensitive systems behind controlled tools with policy enforcement
Explanation:
Data residency and PII minimization are first-class architecture constraints: approved regions, least-privilege tools, and careful prompt contents. Cheapest global routing, full CRM dumps, and unencrypted indefinite storage violate residency and privacy design.

2. Which artifacts best support handoff from architecture to engineering for a Claude system?

A. Only a marketing one-pager
B. ADRs, sequence diagrams, tool contracts, prompt/policy versions, eval plans, runbooks, and environment config references
C. A single verbal hallway conversation
D. An undocumented demo environment

Correct:
ADRs, sequence diagrams, tool contracts, prompt/policy versions, eval plans, runbooks, and environment config references
Explanation:
Handoffs need durable artifacts: ADRs, diagrams, contracts, versions, evals, runbooks, and config references. Verbal-only, marketing-only, or undocumented demos fail engineering handoff.

3. A business unit wants to reuse another unit's Claude agent. What lifecycle concern is most important?

A. Copying the system prompt into email and declaring done
B. Sharing production API keys in a wiki
C. Revalidating data access, policies, evals, and SLAs for the new context before reuse
D. Assuming identical data sensitivity and SLAs without review

Correct:
Revalidating data access, policies, evals, and SLAs for the new context before reuse
Explanation:
Reuse requires revalidation of access, policies, evals, and SLAs in the new context. Assuming sameness, emailing prompts as the whole transfer, or wiki keys are failures.

4. Which is the best description of an evaluator-optimizer pattern in Claude architectures?

A. Always use five agents for every user message
B. Two models argue randomly until one times out
C. Replace evaluation with higher temperature sampling
D. A generator produces a candidate; an evaluator scores it against criteria and the system iterates improve/reject until thresholds are met or a budget is exhausted

Correct:
A generator produces a candidate; an evaluator scores it against criteria and the system iterates improve/reject until thresholds are met or a budget is exhausted
Explanation:
Evaluator-optimizer pairs generation with explicit evaluation criteria and iterative improvement under a budget. It is a controlled quality loop, not random arguing, fixed agent counts, or temperature-based 'evaluation.'

5. What is a sound approach to chunking documents for enterprise RAG?

A. Random 10-character chunks only
B. Chunk with overlap/structure awareness (headings, semantic boundaries), store metadata (source, access labels), and tune chunk size for retrieval quality
C. One giant chunk per company forever
D. Chunk only images and discard text

Correct:
Chunk with overlap/structure awareness (headings, semantic boundaries), store metadata (source, access labels), and tune chunk size for retrieval quality
Explanation:
Effective chunking respects document structure, uses sensible sizes with overlap when needed, and preserves metadata for filtering and citations. Giant single chunks, tiny random shards, or text discard degrade retrieval.

6. What is a key risk of stuffing retrieved chunks into context without ranking or budget limits?

A. It guarantees GDPR compliance
B. Noise, distraction, higher cost/latency, and potential contradiction among chunks can degrade answer quality
C. It always improves accuracy without trade-offs
D. It disables the model completely

Correct:
Noise, distraction, higher cost/latency, and potential contradiction among chunks can degrade answer quality
Explanation:
Unfiltered chunk stuffing increases noise, cost, and latency and can introduce contradictions. It does not always improve accuracy and has nothing to do with guaranteeing compliance.

7. What is the primary purpose of putting stable role, policy, and output-format rules in the system prompt?

A. To provide authoritative, reusable instructions that constrain behavior across turns more reliably than restating rules only in user messages
B. To increase temperature automatically
C. To disable tool use
D. To bypass safety systems

Correct:
To provide authoritative, reusable instructions that constrain behavior across turns more reliably than restating rules only in user messages
Explanation:
System prompts hold stable, authoritative instructions-role, policies, formats-reused across turns and weighted as system-level guidance. They do not set temperature, do not inherently disable tools, and must not be used to attempt safety bypass.

8. A startup wants to ship a Claude feature in two weeks. Stakeholders demand perfect multi-agent automation on day one. What should the architect recommend?

A. Wait two years until a perfect platform exists
B. Outsource the entire architecture to an unreviewed vendor prompt pack
C. Deliver a thinner vertical slice with clear success metrics and evals, then incrementally add agent complexity only if metrics require it
D. Ship the multi-agent system without evaluation because speed matters more

Correct:
Deliver a thinner vertical slice with clear success metrics and evals, then incrementally add agent complexity only if metrics require it
Explanation:
Professional delivery favors a thin, evaluated vertical slice that ships value safely, then adds complexity only when metrics justify it. Shipping multi-agent without evals is reckless. Waiting indefinitely or using unreviewed black-box prompts fails architecture ownership.

9. A team wants custom slash commands and shared project memory for Claude Code. Where should durable team conventions live?

A. In versioned project configuration such as CLAUDE.md / shared settings, reviewed like other code
B. Only in ephemeral chat messages
C. Hardcoded into a single developer's shell history
D. In a sticky note on one monitor

Correct:
In versioned project configuration such as CLAUDE.md / shared settings, reviewed like other code
Explanation:
Team conventions belong in versioned project memory/config (e.g., CLAUDE.md and shared settings) reviewed like code. Ephemeral chat, sticky notes, and personal shell history do not scale.

10. A model refuses a benign support request due to over-refusal. What is the architect's best governance-aware response?

A. Ignore user impact because refusals prove safety
B. Route all refused traffic to an unrestricted open model with no controls
C. Disable all safety systems globally
D. Tune policies and evals to reduce false refusals while preserving true-positive safety blocks; measure both harm rate and over-refusal rate

Correct:
Tune policies and evals to reduce false refusals while preserving true-positive safety blocks; measure both harm rate and over-refusal rate
Explanation:
Safety systems should balance false positives and false negatives via policy tuning and dual metrics. Global disablement or unrestricted fallbacks reintroduce risk; ignoring UX is not sustainable governance.

11. An architecture decision record (ADR) for a Claude system should primarily capture which of the following?

A. A list of employees who like Claude
B. Unstructured chat logs from the design workshop only
C. Only the final prompt text with no rationale
D. Context, options considered, decision, consequences, and constraints (cost, latency, compliance)

Correct:
Context, options considered, decision, consequences, and constraints (cost, latency, compliance)
Explanation:
ADRs document why an architecture was chosen: context, alternatives, decision, consequences, and constraints. Prompt text alone is insufficient. Preference surveys and raw chat logs are not architecture decisions.

12. You must design a claims triage system that classifies severity, extracts structured fields, and routes complex cases to human adjusters. Which high-level pattern best matches this requirement set?

A. Pure single-shot completion with no tools or routing
B. A workflow/routing design with structured extraction, classification, and human-in-the-loop handoff for complex cases
C. An unrestricted autonomous agent that pays claims without human review
D. Batch fine-tuning only, with no runtime Claude calls

Correct:
A workflow/routing design with structured extraction, classification, and human-in-the-loop handoff for complex cases
Explanation:
Claims triage needs classification, structured extraction, routing, and human handoff for complex cases-classic workflow/routing architecture with human-in-the-loop. Pure single-shot lacks routing and structured process control. Fully autonomous payment is inappropriate for regulated claims. Fine-tuning alone does not implement runtime triage workflows.

13. When decommissioning a Claude application, which steps are essential?

A. Delete all audit logs immediately against policy
B. Announce nothing and hope users leave
C. Revoke credentials, archive required logs per policy, communicate sunset, migrate or delete data, and update dependency inventories
D. Leave API keys active forever just in case

Correct:
Revoke credentials, archive required logs per policy, communicate sunset, migrate or delete data, and update dependency inventories
Explanation:
Decommissioning revokes access, handles data/logs per policy, communicates sunset, and updates inventories. Eternal keys, illegal log deletion, and silent abandonment create risk.

14. What should status updates to stakeholders during a multi-month Claude program emphasize?

A. Daily screenshots of the model logo
B. Internal team conflicts without remediation plans
C. Only lines of prompt text written
D. Progress against outcomes and risks: metrics trends, milestones, blockers, decisions needed, and residual risk changes

Correct:
Progress against outcomes and risks: metrics trends, milestones, blockers, decisions needed, and residual risk changes
Explanation:
Program status should track outcomes, metrics, milestones, blockers, decisions, and risk -not vanity activity, logos, or unmanaged conflict reports.

15. Why might an architect use Sonnet for a complex tool-using agent loop rather than Haiku?

A. Sonnet generally offers stronger multi-step reasoning and tool-use reliability for complex loops, justifying higher per-call cost when quality matters
B. Haiku only works offline
C. Haiku cannot call tools under any configuration
D. Sonnet is always free

Correct:
Sonnet generally offers stronger multi-step reasoning and tool-use reliability for complex loops, justifying higher per-call cost when quality matters
Explanation:
Sonnet is commonly preferred for complex agentic tool use because of stronger multi-step reasoning and reliability versus Haiku. Haiku can use tools but may underperform on hard loops. Sonnet is not free, and Haiku is not offline-only.

16. Which handoff package is most appropriate when transferring a Claude agent from a vendor pilot to internal platform ownership?

A. A thank-you note without technical artifacts
B. Code/config repos, prompt versions, eval suites, data maps, threat model, runbooks, SLAs, and known issues list
C. Production traffic cutover with no documentation
D. Access passwords only in a text message

Correct:
Code/config repos, prompt versions, eval suites, data maps, threat model, runbooks, SLAs, and known issues list
Explanation:
Ownership transfer needs complete technical and operational artifacts. Passwords in texts, thank-you notes, or doc-free cutovers create severe operational risk.

17. A team proposes an orchestrator agent that delegates research, drafting, and compliance review to worker agents for a regulated report. When is multi-agent orchestration justified over a single well-prompted Claude call with tools?

A. Only when specialized roles, independent evaluation, or parallel workstreams clearly improve quality or risk control beyond a single agent
B. Whenever the prompt exceeds 1,000 tokens
C. Whenever more than one tool is needed
D. Always, because multi-agent systems are the Professional standard

Correct:
Only when specialized roles, independent evaluation, or parallel workstreams clearly improve quality or risk control beyond a single agent
Explanation:
Anthropic's architecture guidance favors the simplest design that works. Multi-agent orchestration is justified when distinct roles (research, draft, compliance), independent evaluation, or parallelization improve quality, safety, or throughput in ways a single agent cannot. Tool count or prompt length alone does not require multi-agent design, and multi-agent is not automatically 'more professional.'

18. A user attempts to extract another tenant's data via prompt injection. Which control is most decisive?

A. A polite system prompt asking users not to do that
B. Tenant-aware authorization enforced in retrieval and tools so unauthorized data cannot be fetched even if the model is tricked
C. Relying solely on the model to recall who is authorized
D. Hiding the tenant ID in CSS

Correct:
Tenant-aware authorization enforced in retrieval and tools so unauthorized data cannot be fetched even if the model is tricked
Explanation:
Authorization must be enforced in data/tools layers so injection cannot expand access. Soft prompts, CSS hiding, and model memory are not access control.

19. How should an architect handle tool results that contain untrusted HTML or third-party text before they re-enter the model context?

A. Sanitize/delimit them as untrusted data, apply size limits, and instruct the model not to follow instructions found inside tool results
B. Disable all tools permanently
C. Treat them as equally authoritative as system policy
D. Execute any scripts found in tool results immediately

Correct:
Sanitize/delimit them as untrusted data, apply size limits, and instruct the model not to follow instructions found inside tool results
Explanation:
Tool results can carry indirect prompt injection. Architectures should delimit/sanitize, limit size, and instruct that tool content is data, not authority. Elevating tool text to system authority or executing scripts is dangerous; disabling all tools is unnecessary if controls exist.

20. An enterprise needs Claude to access both public product docs and private customer contracts. What integration control is essential?

A. Separate corpora or row-level security with user/tenant-aware retrieval filters so private contracts are only retrieved for authorized principals
B. Put private contracts in the system prompt for all users
C. Publish contracts to the public docs CDN for simpler RAG
D. Mix all documents in one unauthenticated index

Correct:
Separate corpora or row-level security with user/tenant-aware retrieval filters so private contracts are only retrieved for authorized principals
Explanation:
Authorization-aware retrieval is mandatory: separate indexes or filtered retrieval by tenant/user so private contracts never surface to unauthorized users. Shared unauthenticated indexes and global system-prompt inclusion leak data.

21. What should a regression gate in Cl do when prompt changes fail golden evaluations?

A. Delete failing cases from the suite
B. Block or require explicit risk acceptance with documented waiver and owner
C. Silently lower thresholds until green
D. Auto-merge anyway to maintain velocity

Correct:
Block or require explicit risk acceptance with documented waiver and owner
Explanation:
Failed golden evals should block merge or require explicit, documented risk acceptance. Auto-merge, silent threshold cuts, and deleting failing cases undermine the gate.

22. What is the best use of stop sequences or explicit output delimiters in production prompts?

A. To bound generation into parseable regions and reduce trailing chatter that breaks parsers
B. To increase token billing artificially
C. To replace authentication
D. To make the model ignore safety systems

Correct:
To bound generation into parseable regions and reduce trailing chatter that breaks parsers
Explanation:
Stop sequences and delimiters help produce bounded, parseable outputs and reduce trailing noise that breaks downstream parsers. They are not safety bypasses, billing tools, or auth mechanisms.

23. Which platform capability most improves developer productivity for Claude integrations across teams?

A. Forcing every team to rebuild auth, logging, and MCP connectors from scratch
B. Hiding documentation behind undocumented tribal knowledge
C. Shared SDKs, golden paths, reusable MCP servers/skills, templates, and self-serve observability
D. A 40-page PDF with no runnable examples

Correct:
Shared SDKs, golden paths, reusable MCP servers/skills, templates, and self-serve observability
Explanation:
Platforms multiply productivity via shared SDKs, golden paths, reusable MCP/skills, templates, and observability. Rebuild-everything, inert PDFs, and tribal knowledge slow teams.

24. A pilot is successful with 20 users. Leadership wants company-wide launch next week. What should the architect communicate?

A. Scaling needs capacity plans, support model, training, eval coverage for new intents, access control reviews, and phased rollout criteria
B. That no communication is needed
C. That security review can wait until after full launch
D. That pilot success automatically scales without further work

Correct:
Scaling needs capacity plans, support model, training, eval coverage for new intents, access control reviews, and phased rollout criteria
Explanation:
Pilot-to-scale requires capacity, support, training, broader evals, access reviews, and phased criteria. Automatic scale assumptions and post-launch security are reckless.

25. A stakeholder says 'we need an agent.' What should you do first?

A. Translate the request into jobs-to-be-done, constraints, and measurable outcomes; then choose the simplest pattern that fits
B. Promise full autonomy with no human oversight
C. Spin up multi-agent infrastructure the same day
D. Buy GPUs before requirements exist

Correct:
Translate the request into jobs-to-be-done, constraints, and measurable outcomes; then choose the simplest pattern that fits
Explanation:
Stakeholder language like 'agent' must be translated into jobs, constraints, and metrics before pattern selection. Infrastructure-first, GPU-first, or autonomy promises without oversight are premature.

26. An enterprise wants a Claude solution that answers HR policy questions from a curated knowledge base with citations. Latency must stay under 3 seconds and answers must be grounded. Which architecture is the most appropriate starting point?

A. An open-ended agent with unrestricted web browsing as the only knowledge source
B. Fine-tune a custom model on HR policies before any retrieval layer
C. A multi-agent swarm with specialized agents negotiating every answer
D. A RAG-augmented single-turn or short workflow design with retrieval, grounding instructions, and citation requirements

Correct:
A RAG-augmented single-turn or short workflow design with retrieval, grounding instructions, and citation requirements
Explanation:
For curated, citable knowledge Q&A with a tight latency budget, a RAG-augmented design is the right default: retrieve relevant policy chunks, constrain the model to ground answers in those chunks, and require citations. Multi-agent negotiation and unrestricted browsing add latency and risk without improving grounding for a fixed corpus. Fine-tuning alone does not guarantee up-to-date, citable retrieval and is usually premature before a retrieval baseline works.

27. For a long-running investigation workflow that may take hours, which orchestration approach is most appropriate?

A. Keep a single synchronous HTTP request open for hours without checkpoints
B. Use durable workflow orchestration with persisted state, resumable steps, tool results stored securely, and user-visible status
C. Restart from scratch on every browser refresh with no state
D. Store intermediate secrets in client-side localStorage only

Correct:
Use durable workflow orchestration with persisted state, resumable steps, tool results stored securely, and user-visible status
Explanation:
Long-running work needs durable orchestration: persisted state, resumable steps, secure storage of intermediate results, and status visibility. Multi-hour synchronous HTTP is fragile. Stateless restarts waste work. Client-only secret storage is insecure.

28. A team only reviews model quality by 'vibes' in a chat window. What should the architect introduce?

A. An eval harness with offline metrics, sampled human review, and Cl gates on prompt/agent changes
B. Replacing metrics with higher temperature
C. Disabling all testing to ship faster
D. More vibes from more people

Correct:
An eval harness with offline metrics, sampled human review, and Cl gates on prompt/agent changes
Explanation:
Professional evaluation uses harnesses, metrics, sampled human review, and Cl gates. Vibes-only, no testing, or temperature changes are not evaluation frameworks.

29. Why include adversarial and jailbreak-style cases in eval sets for enterprise assistants?

A. To measure resilience of policy and safety controls under hostile or tricky inputs before attackers find them in production
B. To replace unit tests for backend billing
C. Because they improve marketing copy creativity
D. To entertain the team

Correct:
To measure resilience of policy and safety controls under hostile or tricky inputs before attackers find them in production
Explanation:
Adversarial cases stress safety and policy controls proactively. They are not entertainment, marketing tools, or billing unit tests.

30. A team uses temperature O for extraction and temperature 0.8 for creative marketing variants. What principle does this illustrate?

A. Sampling parameters should match task goals: lower for deterministic extraction, higher for diverse creative outputs
B. Higher temperature always improves factual accuracy
C. Temperature replaces the need for evaluation
D. Temperature must be identical across the enterprise

Correct:
Sampling parameters should match task goals: lower for deterministic extraction, higher for diverse creative outputs
Explanation:
Temperature is task-dependent: low for deterministic extraction/classification, higher for creative diversity. Uniform enterprise temperature is not required. Temperature does not replace evals and higher temperature usually hurts factual precision.

31. What is the best way to present model limitations to non-technical executives?

A. Explain concrete failure modes, residual risks, human oversight needs, and what the system will not do, tied to business impact
B. Use only research jargon without examples
C. Hide limitations in an appendix nobody reads without a summary
D. Claim zero limitations to secure budget

Correct:
Explain concrete failure modes, residual risks, human oversight needs, and what the system will not do, tied to business impact
Explanation:
Executive communication needs concrete failure modes, residual risks, oversight, and non-goals linked to business impact. Jargon-only, zero-limitation claims, or hidden appendices without summaries fail transparency.

32. Which failure mode is most associated with poor RAG hybrid search design?

A. Automatic compliance certification
B. Missing exact identifiers (ticket IDs, SKUs) when using semantic search alone without keyword/hybrid retrieval
C. Guaranteed prevention of all hallucinations without generation
D. Perfect recall with zero cost always

Correct:
Missing exact identifiers (ticket IDs, SKUs) when using semantic search alone without keyword/hybrid retrieval
Explanation:
Dense semantic search alone often misses exact tokens like IDs and SKUs; hybrid/keyword components fix this common failure. RAG does not magically eliminate cost, hallucination without generation, or compliance certification.

33. How can platform teams reduce unsafe MCP sprawl while still enabling developers?

A. Store MCP secrets in public repos for convenience
B. Block all MCP permanently
C. Provide an approved catalog, rapid review path for new servers, default least-privilege templates, and automated policy checks in Cl
D. Allow any MCP with admin scopes by default

Correct:
Provide an approved catalog, rapid review path for new servers, default least-privilege templates, and automated policy checks in Cl
Explanation:
Enablement with control means an approved catalog, fast review, least-privilege defaults, and Cl policy checks. Total blocks slow innovation; admin-by-default and public secrets are dangerous.

34. Which control best reduces the risk of a Claude agent performing unauthorized financial transfers?

A. Hard authorization checks and transaction limits in the tool layer, dual control for high-value actions, and human approval gates
B. Rely on the model promising not to transfer money
C. Increase temperature so actions are more random
D. Hide the transfer tool name so the model cannot guess it

Correct:
Hard authorization checks and transaction limits in the tool layer, dual control for high-value actions, and human approval gates
Explanation:
Side-effecting financial actions need hard authorization, limits, dual control, and human gates in the tool/application layer-not model promises, obscurity, or randomness.

35. For citation-grade RAG answers, what should the generator be required to do?

A. Cite the system prompt as the document source
B. Ground claims in retrieved evidence and emit citations tied to actual retrieved chunk IDs/sources; refuse or hedge when evidence is insufficient
C. Never mention sources to appear more confident
D. Invent plausible URLs when sources are missing

Correct:
Ground claims in retrieved evidence and emit citations tied to actual retrieved chunk IDs/sources; refuse or hedge when evidence is insufficient
Explanation:
Citation-grade systems must ground claims, cite real retrieved sources, and refuse/hedge without evidence. Invented URLs, suppressed sources, or citing the system prompt as a document are incorrect.

36. An enterprise chat app repeatedly sends the same 8k-token policy manual in every request. Which context-engineering lever should be applied first?

A. Duplicate the manual three more times for emphasis
B. Raise max_tokens for every response to match the manual size
C. Use prompt caching / stable prefix strategies and retrieve only relevant sections instead of full dumps every turn
D. Remove all user messages to free space

Correct:
Use prompt caching / stable prefix strategies and retrieve only relevant sections instead of full dumps every turn
Explanation:
Repeated large static prefixes are ideal for prompt caching and for retrieval of relevant sections rather than full-document dumps. Duplication wastes tokens. Raising max_tokens does not fix input waste. Removing user messages destroys the task.

37. Which statement best describes a production-ready target architecture for an enterprise Claude system?

A. A single prompt file shared over chat with no version control
B. Direct browser access to Claude with no application boundary
C. A notebook demo with hardcoded API keys and no observability
D. Documented components for ingress, auth, orchestration, model/tool access, retrieval, evaluation hooks, logging, and human escalation paths

Correct:
Documented components for ingress, auth, orchestration, model/tool access, retrieval, evaluation hooks, logging, and human escalation paths
Explanation:
Enterprise target architecture includes clear components: secure ingress and auth, orchestration, model and tool access, retrieval if needed, evaluation and logging hooks, and escalation to humans. Notebook demos, unversioned prompts, and browser-only access lack the control planes required for production.

38. How should auditability be designed for agent actions?

A. Record who initiated the session, prompts/policy versions, tool calls, parameters, results, approvals, and model identifiers with tamper-evident storage appropriate to risk
B. Store logs in an editable shared spreadsheet with no access control
C. Avoid logs to reduce storage costs always
D. Log only the final sentence shown to the user

Correct:
Record who initiated the session, prompts/policy versions, tool calls, parameters, results, approvals, and model identifiers with tamper-evident storage appropriate to risk
Explanation:
Auditable agents need end-to-end records of initiator, versions, tools, approvals, and model IDs in protected storage. Minimal final-text logs and editable spreadsheets fail enterprise audit needs.

39. A product owner wants sub-500ms p95 latency for autocomplete-style suggestions, while a separate research assistant can take up to 30 seconds. How should the architect allocate Claude models and patterns?

A. Disable streaming for autocomplete so full answers are buffered
B. Force both workloads through extended thinking with maximum tokens
C. Use a low-latency model and short context path for autocomplete; reserve deeper models, tools, and longer workflows for the research assistant
D. Use the same multi-agent Opus stack for both to maximize quality uniformly

Correct:
Use a low-latency model and short context path for autocomplete; reserve deeper models, tools, and longer workflows for the research assistant
Explanation:

Different SLAs require different architectures. Autocomplete needs Haiku-class (or similarly fast) models, tight prompts, and minimal tooling. Research can use Sonnet/Opus, tools, RAG, and longer agent loops. Uniform multi-agent Opus for both wastes cost and breaks the latency SLA. Extended thinking and full buffering worsen autocomplete latency.

40. What is the best first step to enable a large engineering org with Claude Code safely?

A. Publish baseline standards (CLAUDE.md patterns, permissions, approved MCP servers), provide training, and pilot with guardrails before broad rollout
B. Ban Claude Code to avoid all risk
C. Allow each team to invent incompatible setups with no platform guidance
D. Give every developer unrestricted production credentials on day one

Correct:
Publish baseline standards (CLAUDE.md patterns, permissions, approved MCP servers), provide training, and pilot with guardrails before broad rollout
Explanation:
Org enablement needs standards, training, approved integrations, and guarded pilots. Unrestricted prod credentials, total bans, or zero guidance all fail productivity-with-safety goals.

41. Which safety layering approach is most appropriate for user-facing Claude apps?

A. Only a single system-prompt sentence saying 'be safe'
B. No safety because Claude is always aligned for every enterprise policy
C. Defense in depth: policy prompts, input/output filters, tool authorization, rate limits, monitoring, and escalation paths
D. Safety only at the CDN layer

Correct:
Defense in depth: policy prompts, input/output filters, tool authorization, rate limits, monitoring, and escalation paths
Explanation:
Safety needs defense in depth across prompt policy, filters, tool auth, rate limits, monitoring, and escalation. A single prompt line, CDN-only controls, or assuming perfect alignment are insufficient for enterprise policy.

42. How should cost be treated in Professional Claude solution design?

A. Model token, tool, retrieval, and human-review costs against expected volume and set guardrails (caching, model tiering, max tokens, sampling rates)
B. Ignore cost until after go-live
C. Charge users nothing and never measure usage
D. Always use the largest model with maximum output tokens

Correct:
Model token, tool, retrieval, and human-review costs against expected volume and set guardrails (caching, model tiering, max tokens, sampling rates)
Explanation:
Cost is an explicit design constraint: estimate unit economics, apply caching and model tiering, cap tokens, and sample expensive paths. Ignoring cost or always maxing model size breaks enterprise viability. Not measuring usage makes cost control impossible.

43. An architect is comparing prompt chaining versus a single long agent loop for a multi-step document analysis pipeline with stable, sequential stages. Which choice is usually better and why?

A. Prefer prompt chaining or a structured workflow when stages are stable and independently testable; reserve open agent loops for open-ended exploration
B. Always use a free-form agent loop because chaining is obsolete
C. Use multi-agent negotiation for every document stage
D. Skip intermediate stages and ask one model call for the final answer only

Correct:
Prefer prompt chaining or a structured workflow when stages are stable and independently testable; reserve open agent loops for open-ended exploration
Explanation:
When stages are known and stable (extract → analyze summarize format), prompt chaining or a structured workflow improves testability, observability, and failure isolation. Open agent loops fit less predictable exploration. Multi-agent negotiation is usually unnecessary for fixed sequential stages. Skipping stages often reduces accuracy and auditability.

44. A client asks for 'an agent that can do anything in our SAP landscape.' What is the architect's best first response in solution design?

A. Grant the agent admin credentials to SAP immediately
B. Decompose into scoped use cases, data boundaries, allowed tools, success metrics, and human approval gates before designing autonomy
C. Build a single mega-prompt that lists every SAP transaction
D. Refuse any SAP integration on principle

Correct:
Decompose into scoped use cases, data boundaries, allowed tools, success metrics, and human approval gates before designing autonomy
Explanation:
Professional architecture starts by scoping: which tasks, which systems, which tools, which data, which metrics, and where humans approve. Broad 'do anything' agents with admin credentials are unsafe and untestable. Mega-prompts are not a substitute for tool and authorization design. Blanket refusal is unnecessary when scoped integrations are feasible.

45. How should few-shot examples be chosen for a production classifier prompt?

A. Random unrelated examples from the internet
B. As many examples as possible until the context window is full every time
C. Only adversarial examples with no normal cases
D. Representative, labeled examples covering common and edge cases, versioned and tested for prompt regressions

Correct:
Representative, labeled examples covering common and edge cases, versioned and tested for prompt regressions
Explanation:
Few-shot examples should be curated, representative, cover edges, and be version-controlled with regression tests. Random web examples, always-maxing context, or only adversarial cases degrade quality and cost control.

46. A backend service needs server-to-server Claude calls with strict service identity and no interactive user. Which integration style fits best?

A. Browser-only session cookies shared among servers
B. Server-side Claude API integration using service credentials stored in a secret manager, with least-privilege IAM
C. Embedding API keys in mobile apps
D. Emailing prompts to a shared inbox for manual processing

Correct:
Server-side Claude API integration using service credentials stored in a secret manager, with least-privilege IAM
Explanation:
Server-to-server workloads should call the Claude API with service credentials from a secret manager under least privilege. Shared browser cookies, mobile-embedded keys, and email queues are insecure or non-viable for this pattern.

47. A pipeline must emit machine-readable JSON that downstream services validate against a schema. What is the best prompting/architecture approach?

A. Use structured outputs / tool-based schema constraints, validate responses, and define repair or fail paths on schema errors
B. Accept any string and cast blindly to objects
C. Increase temperature to encourage creative field names
D. Ask for 'JSON-ish' free text and parse with regex hope

Correct:
Use structured outputs / tool-based schema constraints, validate responses, and define repair or fail paths on schema errors
Explanation:
Production structured data needs schema-constrained outputs (tools/structured outputs), validation, and defined repair/fail behavior. Regex hope, blind casting, and high temperature for field names create fragile integrations.

48. An A/B test shows version B improves user thumbs-up by 3% but increases hallucinations on a golden legal set. What should the architect recommend?

A. Ship B immediately because thumbs-up is the only metric
B. Do not ship B to legal flows; investigate metric conflict, segment traffic, and prioritize safety/golden-set quality for high-stakes paths
C. Delete the golden set so the conflict disappears
D. Increase temperature on B to improve thumbs-up further

Correct:
Do not ship B to legal flows; investigate metric conflict, segment traffic, and prioritize safety/golden-set quality for high-stakes paths
Explanation:
Conflicting metrics require segmented decisions: engagement gains cannot override safety regressions on legal flows. Deleting golden sets or raising temperature worsens governance.

49. Which operational practice best supports continuous improvement of a Claude platform used by many product teams?

A. Measure only the number of Slack messages containing the word Claude
B. Never collect feedback after launch
C. Force a single weekly all-hands rewrite of every prompt company-wide
D. Run office hours, track common failure themes, publish golden-path updates, and measure adoption, time-to-first-success, and incident rates

Correct:
Run office hours, track common failure themes, publish golden-path updates, and measure adoption, time-to-first-success, and incident rates
Explanation:
Platform Ops improves via feedback loops, themed fixes, golden-path updates, and metrics like adoption, time-to-first-success, and incidents. No feedback, chaotic rewrites, or vanity Slack counts do not improve the platform.

50. What should a production runbook for a Claude service include?

A. Only the CEO biography
B. A joke list for chats
C. On-call contacts, failure modes, dashboards, rollback steps, prompt/version pins, dependency status, and escalation paths
D. Credentials in plaintext at the top

Correct:
On-call contacts, failure modes, dashboards, rollback steps, prompt/version pins, dependency status, and escalation paths
Explanation:
Runbooks cover who to call, what fails, where to look, how to roll back, which versions are live, dependencies, and escalations. Biographies, jokes, and plaintext credentials do not belong.

51. Choosing stdio vs HTTP transport for an MCP server in enterprise settings mainly depends on what?

A. stdio always being required for multi-tenant SaaS
B. Whether Claude can spell HTTP
C. Whether the brand color is blue
D. Process locality and deployment model: local/dev co-process vs remote/shared network service with operational controls

Correct:
Process locality and deployment model: local/dev co-process vs remote/shared network service with operational controls
Explanation:
stdio suits local co-process integrations; HTTP suits remote/shared services with network controls. Brand color is irrelevant. Multi-tenant SaaS usually needs remote HTTP-style deployments, not local stdio for each user.

52. When is extended thinking most justified in a Professional architecture?

A. For every autocomplete keystroke
B. To hide intermediate tool failures from logs
C. For hard multi-step reasoning or planning tasks where quality gains outweigh latency and cost, with budgets and evaluation in place
D. As a substitute for retrieval when documents are unavailable

Correct:
For hard multi-step reasoning or planning tasks where quality gains outweigh latency and cost, with budgets and evaluation in place
Explanation:
Extended thinking helps difficult reasoning/planning when quality gains justify latency and cost, under explicit budgets and evals. It is wrong for ultra-low-latency paths, is not a retrieval substitute, and must not hide operational failures.

53. Which practice best keeps architecture decisions aligned as requirements evolve?

A. Never revisit ADRs once written
B. Delete history so only the latest prompt exists
C. Allow each engineer to fork production architecture privately
D. Revisit ADRs on trigger events (metric misses, new regulations, major volume changes) and record superseding decisions

Correct:
Revisit ADRs on trigger events (metric misses, new regulations, major volume changes) and record superseding decisions
Explanation:
Living architecture revisits ADRs on material triggers and records supersessions. Freezing forever, deleting history, or private forks undermine lifecycle control.

54. For a HIPAA-relevant healthcare assistant architecture (high-level), which practice is most aligned?

A. Store PHI in plaintext prompts in shared tickets forever
B. Minimize PHI in prompts, use approved environments/BAAs as required, access controls, audit logs, and avoid unnecessary retention of sensitive model I/O
C. Email PHI to personal Gmail for faster debugging
D. Disable audit logging to protect privacy

Correct:
Minimize PHI in prompts, use approved environments/BAAs as required, access controls, audit logs, and avoid unnecessary retention of sensitive model I/O
Explanation:
HIPAA-oriented design minimizes PHI exposure, uses approved environments and contracts, enforces access control and auditability, and limits retention. Plaintext forever stores, personal email, and disabled audits are violations of good practice.

55. When designing capacity for a Claude-backed API that spikes at 9am daily, what should the architect plan for first?

A. Unlimited retries with no backoff
B. Peak concurrency, token throughput, queueing/backpressure, timeouts, and fallback behavior under rate limits
C. Only average QPS, ignoring peaks
D. Disabling caching to force fresh model calls for every request

Correct:
Peak concurrency, token throughput, queueing/backpressure, timeouts, and fallback behavior under rate limits
Explanation:
Capacity design must account for peak concurrency and token throughput, plus queueing, backpressure, timeouts, and graceful degradation when rate-limited. Average-only sizing fails at peaks. Unlimited retries amplify outages. Disabling caching increases load and cost without improving peak resilience.

56. Which prompting practice best reduces prompt-injection impact in a tool-using enterprise assistant?

A. Concatenate user text into the system prompt as authority
B. Put untrusted user content and tool outputs in clearly delimited sections and instruct the model that system policies override them
C. Trust tool outputs more than system policy by default
D. Disable all system instructions

Correct:
Put untrusted user content and tool outputs in clearly delimited sections and instruct the model that system policies override them
Explanation:
Defense-in-depth for injection includes delimiting untrusted content, keeping policies in the system layer, and instructing that system rules override user/tool text. Elevating user or tool text into system authority increases injection risk.

57. What observability should Claude Code / agent Cl pipelines expose to developers?

A. Nothing, to keep pipelines mysterious
B. Eval results, token/cost estimates, failed cases, permission diffs, and links to traces for failing scenarios
C. Raw production customer PII dumps in public Cl logs
D. Only a green checkmark with no artifacts

Correct:
Eval results, token/cost estimates, failed cases, permission diffs, and links to traces for failing scenarios
Explanation:
Developer-useful Cl shows eval outcomes, cost signals, failures, permission diffs, and traces-not mystery, empty greens, or PII in public logs.

58. What is a primary benefit of defining tool error responses in a structured way for agents?

A. It confuses the model on purpose
B. It removes the need for timeouts
C. It allows the agent to recover, retry, or escalate with actionable error codes/messages instead of failing opaquely
D. It makes authentication unnecessary

Correct:
It allows the agent to recover, retry, or escalate with actionable error codes/messages instead of failing opaquely
Explanation:
Structured tool errors enable recovery, alternative strategies, or escalation. They do not remove timeouts or authentication and should not intentionally confuse the model.

59. Which metric set best monitors a production Claude API integration?

A. Request rate, error rate, latency percentiles, token usage/cost, tool success rates, and groundedness/quality samples
B. Number of employees who like the logo
C. Only CSS page load time
D. Git commit count alone

Correct:
Request rate, error rate, latency percentiles, token usage/cost, tool success rates, and groundedness/quality samples
Explanation:
Operational monitoring for Claude integrations includes traffic, errors, latency, tokens/cost, tool outcomes, and sampled quality/groundedness. Frontend CSS, logo popularity, and commit counts are not sufficient integration SLOS.

60. An architect needs Claude to follow a multi-constraint style guide (tone, banned phrases, length, locale). Which approach is most robust?

A. Rely solely on the model remembering verbal feedback from last month
B. Hide constraints in tool names only
C. Mention constraints casually once in a user message only
D. Encode constraints clearly in system/developer instructions, provide examples, validate outputs with automated checks, and repair or reject failures

Correct:
Encode constraints clearly in system/developer instructions, provide examples, validate outputs with automated checks, and repair or reject failures
Explanation:
Multi-constraint style adherence needs clear system instructions, examples, automated validators, and repair/reject paths. Casual one-off user mentions and long-term 'memory' of verbal feedback are unreliable. Tool names cannot encode full style policy.

61. Which statement correctly contrasts MCP tools vs MCP resources?

A. Tools are side-effecting or computational actions; resources are readable contextual data the model can fetch
B. Resources replace authentication for tools
C. They are identical concepts with different marketing names
D. Resources execute writes; tools are read-only by definition always

Correct:
Tools are side-effecting or computational actions; resources are readable contextual data the model can fetch
Explanation:
In MCP, tools typically perform actions/computations (often with side effects), while resources provide readable context/data. They are not identical, and resources do not replace authentication.

62. What is residual risk acceptance in Al governance?

A. Pretending residual risk is zero after launch
B. Ignoring risks that are hard to measure
C. Transferring all blame to the model vendor without analysis
D. Documenting remaining risks after controls, with an accountable owner who accepts them for a defined scope and review period

Correct:
Documenting remaining risks after controls, with an accountable owner who accepts them for a defined scope and review period
Explanation:
Residual risk acceptance is explicit: document what remains after controls, who accepts it, for what scope, and when it is reviewed. Denial, pure blame-shift, or ignoring hard risks is not acceptance.

63. When should an architect typically prefer Claude Haiku over Claude Opus in a multi-step system?

A. For the final legal sign-off on a high-stakes contract with no review
B. For high-volume, low-latency classification, routing, or simple extraction stages where capability needs are modest
C. Whenever accuracy is more important than cost, regardless of task
D. Only when the user explicitly types the word Haiku

Correct:
For high-volume, low-latency classification, routing, or simple extraction stages where capability needs are modest
Explanation:
Haiku is the right default for high-volume, latency-sensitive, simpler stages (classify, route, extract). High-stakes legal sign-off without review should not rely on the smallest model. 'Always accuracy first' ignores task fit. Model choice is an architecture decision, not a user keyword.

64. A Claude agent must create a support ticket. Which tool design is best?

A. Hardcode ticket creation in the client with no model tool
B. A vague tool named do_stuff with a free-form string argument and no schema
C. Allow the model to SSH into production and run SQL freely
D. A clearly named create_support_ticket tool with typed parameters, validation, idempotency key, and documented side effects

Correct:
A clearly named create_support_ticket tool with typed parameters, validation, idempotency key, and documented side effects
Explanation:
Good tools are narrowly named, schema-typed, validated, and explicit about side effects and idempotency. Vague free-form tools and raw production SSH/SQL are dangerous. Skipping tools may be fine for pure UI flows, but if the agent must act, design the tool properly.

65. What is the role of inter-rater reliability when using human judges for LLM outputs?

A. It guarantees model factuality automatically
B. It is irrelevant for subjective tasks
C. It measures agreement among reviewers and indicates whether rubrics are clear enough for consistent scoring
D. It replaces the need for any automated metrics

Correct:
It measures agreement among reviewers and indicates whether rubrics are clear enough for consistent scoring
Explanation:
Inter-rater reliability shows whether humans apply rubrics consistently. Low agreement means rubrics/training need work. It does not auto-guarantee factuality or fully replace automated metrics.

66. A team wants to optimize for both answer completeness and brevity. How should success be specified?

A. Assume completeness and brevity never conflict
B. Optimize only for maximum tokens generated
C. Define explicit rubrics/SLAs (e.g., required fields present, max length, user task completion) and measure each dimension
D. Tell the model 'be perfect' with no measurable definition

Correct:
Define explicit rubrics/SLAs (e.g., required fields present, max length, user task completion) and measure each dimension
Explanation:
Multi-objective quality needs explicit rubrics and measurements per dimension. Vague perfection, max tokens, or assuming no conflict fail optimization practice.

67. At an architecture level, what does GDPR primarily require teams to consider for Claude systems processing personal data?

A. Only the color of consent buttons
B. That encryption is forbidden
C. Lawful basis, purpose limitation, data minimization, retention limits, data subject rights, and appropriate security-including what personal data enters prompts, logs, and training-like stores
D. That all personal data must be posted publicly for transparency

Correct:
Lawful basis, purpose limitation, data minimization, retention limits, data subject rights, and appropriate security-including what personal data enters prompts, logs, and training-like stores
Explanation:
GDPR-oriented architecture addresses lawful basis, purpose, minimization, retention, rights, and security-including prompts and logs. Public posting of personal data and forbidding encryption are wrong.

68. How should tool authentication be designed when Claude agents call internal APIs?

A. Disable auth for internal tools because the agent is trusted
B. Pass the user's plaintext password in the prompt
C. Use delegated, scoped tokens or service identities with least privilege, short TTL, and audit logging-never put long-lived secrets in prompts
D. Share one global admin token across all tools and tenants

Correct:
Use delegated, scoped tokens or service identities with least privilege, short TTL, and audit logging-never put long-lived secrets in prompts
Explanation:
Tools should use scoped, short-lived credentials or proper service identities with audit logs. Secrets must not live in prompts. Global admin tokens and disabled internal auth violate least privilege.

69. Which statement best reflects responsible disclosure and abuse response for a Claude product?

A. Ignore external security reports to protect reputation
B. Provide a channel for reports, triage severity, remediate, communicate as appropriate, and improve detections
C. Publicly shame reporters immediately
D. Turn the product off permanently after any report

Correct:
Provide a channel for reports, triage severity, remediate, communicate as appropriate, and improve detections
Explanation:
Responsible programs accept reports, triage, fix, communicate appropriately, and improve detections. Ignoring reports, shaming reporters, or permanent shutdown for any report are poor responses.

70. How should third-party MCP servers be governed before enterprise use?

A. Review provenance, permissions requested, data flows, security posture, and approve via a platform allowlist with least privilege
B. Grant each server domain-wide admin by default
C. Install any public MCP server with full credentials immediately
D. Trust all open-source code without review because it is popular

Correct:
Review provenance, permissions requested, data flows, security posture, and approve via a platform allowlist with least privilege
Explanation:
Third-party MCP servers need security review, least privilege, and allowlisting. Blind install, popularity-based trust, and default admin rights are unsafe.

71. A latency SLA is p * 95 < 2s . How should the team validate it?

A. Use average latency only and assume p95 matches
B. Ignore tools latency because it is external
C. Load test representative traffic mix, measure percentiles end-to-end (including retrieval/tools), and track production p*95 with alerts
D. Test only on an empty staging system with one request

Correct:
Load test representative traffic mix, measure percentiles end-to-end (including retrieval/tools), and track production p95 with alerts
Explanation:
SLA validation needs representative load tests and production percentile monitoring including dependencies. Single idle requests, ignoring tools, and average-only views misrepresent p95.

72. How should product and risk stakeholders jointly decide whether a Claude feature is ready for production?

A. Wait for perfect zero-risk certainty that never arrives
B. Ship if any single engineer votes yes
C. Ship if the demo is impressive regardless of evals
D. Use a go/no-go checklist covering eval thresholds, security review, privacy assessment, support readiness, and rollback plan

Correct:
Use a go/no-go checklist covering eval thresholds, security review, privacy assessment, support readiness, and rollback plan
Explanation:
Go/no-go needs multi-factor checklists: quality, security, privacy, support, rollback. Demo wow, single votes, or impossible zero-risk waits are poor governance.

73. What is shadow testing (dark launch) for a new Claude prompt or agent version?

A. Forcing all users onto the new version immediately
B. Sending production traffic to the new version without showing results to users, comparing offline to the live version
C. Testing only in a slide deck
D. Deleting logs to hide differences

Correct:
Sending production traffic to the new version without showing results to users, comparing offline to the live version
Explanation:
Shadow testing runs the candidate version on real traffic without user-visible impact, enabling comparison to the live path. It is not log deletion, forced cutover, or slide-deck testing.

74. A production dashboard shows rising cost per successful task after a prompt change. What is the best next step?

A. Ignore cost if revenue exists somewhere
B. Analyze token and tool usage by stage, compare to baseline, and roll back or optimize the change with evals confirming quality is preserved
C. Turn off monitoring to stop the alerts
D. Immediately double max_tokens

Correct:
Analyze token and tool usage by stage, compare to baseline, and roll back or optimize the change with evals confirming quality is preserved
Explanation:
Cost regressions need root-cause analysis by stage and either rollback or optimization with quality checks. Ignoring cost, raising max_tokens, or disabling monitors worsens the problem.

75. A conversation approaches the context limit. Which strategy is most architecturally sound?

A. Reset the user identity on every message
B. Summarize older turns, retain critical entities and tool results, keep system policies, and optionally re-retrieve facts as needed
C. Concatenate all history forever without summarization
D. Silently drop the system prompt to free tokens

Correct:
Summarize older turns, retain critical entities and tool results, keep system policies, and optionally re-retrieve facts as needed
Explanation:
Context management near limits should compress history thoughtfully while preserving system policies, critical state, and tool outcomes, and re-retrieve facts when needed. Dropping system policy, unbounded history, or identity resets break the application.

76. How should success metrics be agreed with business stakeholders for a support copilot?

A. Jointly define leading/lagging metrics such as handle time, deflection (with quality guardrails), CSAT, escalation rate, and safety incident rate
B. Only 'Al magic' qualitative feelings
C. Only model perplexity
D. No metrics until a year after launch

Correct:
Jointly define leading/lagging metrics such as handle time, deflection (with quality guardrails), CSAT, escalation rate, and safety incident rate
Explanation:
Business-aligned metrics include operational and quality measures with safety. Vibes, pure perplexity, or delayed metrics fail lifecycle management.

77. A partner system only supports SOAP APIs. Claude tools need clean JSON schemas. What should the integration layer do?

A. Paste WSDL files into every user message
B. Force the model to author raw SOAP envelopes with secrets inline
C. Abandon the integration because Claude cannot work with any enterprise system
D. Place an adapter/facade that translates typed tool calls to SOAP, handles auth, and returns normalized JSON errors/results

Correct:
Place an adapter/facade that translates typed tool calls to SOAP, handles auth, and returns normalized JSON errors/results
Explanation:
Adapters/facades keep tool interfaces clean while translating to legacy SOAP, managing auth and normalizing responses. Forcing the model to craft SOAP with secrets is brittle and unsafe. Abandoning is unnecessary. Pasting WSDLs into user messages is a poor interface.

78. When is Model Context Protocol (MCP) generally a better fit than ad-hoc one-off HTTP tool wrappers?

A. When you want to avoid all authentication forever
B. When you must prevent any tool use
C. When you need a standardized way to expose tools, resources, and prompts to Claude clients and share integrations across apps
D. When the only goal is to increase prompt size

Correct:
When you need a standardized way to expose tools, resources, and prompts to Claude clients and share integrations across apps
Explanation:
MCP standardizes how tools, resources, and prompts are exposed to Claude-compatible clients, improving reuse and consistency across applications. It does not remove the need for auth, is not for inflating prompts, and enables rather than prevents tool use.

79. What is a golden dataset in Claude evaluation practice?

A. A curated, versioned set of inputs with expected outputs or rubrics used to measure quality regressions over time
B. Random logs never reviewed by humans
C. A set of production secrets used as prompts
D. The largest model available

Correct:
A curated, versioned set of inputs with expected outputs or rubrics used to measure quality regressions over time
Explanation:
Golden datasets are curated, versioned cases with expected outputs or scoring rubrics for regression measurement. They are not secret stores, unreviewed noise, or model tiers.

80. What is the best way to handle dual-use risk when an internal agent can generate code?

A. Store production credentials in the system prompt for convenience
B. Ban all code generation permanently
C. Allow unrestricted production deploy permissions from chat
D. Scope tools to non-prod by default, require separate CI/CD and human review for production changes, and monitor for abusive patterns

Correct:
Scope tools to non-prod by default, require separate CI/CD and human review for production changes, and monitor for abusive patterns
Explanation:
Dual-use code agents should be scoped away from direct prod deploy, with CI/CD and human review for releases plus abuse monitoring. Bans may be unnecessary; prod credentials in prompts are dangerous.

81. Which data retention practice best fits responsible Claude logging?

A. Never log anything, even security events
B. Define retention schedules by data class, redact or tokenize sensitive fields, and restrict access to logs
C. Keep full prompts and tool payloads forever in plaintext for all environments
D. Mirror production PHI logs to public analytics tools

Correct:
Define retention schedules by data class, redact or tokenize sensitive fields, and restrict access to logs
Explanation:
Responsible retention uses class-based schedules, redaction/tokenization, and access control. Forever plaintext, zero logging, and public PHI mirrors are wrong extremes.

82. Which statement about online vs offline evaluation is correct?

A. Online evals never need privacy review
B. Offline evals run on curated sets before release; online evals measure live traffic metrics and experiments after controlled exposure
C. Offline evals use production users as the only test set
D. Offline evals make online monitoring unnecessary forever

Correct:
Offline evals run on curated sets before release; online evals measure live traffic metrics and experiments after controlled exposure
Explanation:
Offline evals gate releases on curated sets; online evals measure real traffic after controlled rollout. Online evals still need privacy review, and offline cannot replace ongoing monitoring.

83. An executive asks for a risk scorecard for a Claude rollout. Which elements should it include?

A. Only the number of demos delivered
B. Threat scenarios, control coverage, residual risk, owners, metrics (safety incidents, quality, privacy), and review cadence
C. Only GPU prices
D. A single green checkbox with no evidence

Correct:
Threat scenarios, control coverage, residual risk, owners, metrics (safety incidents, quality, privacy), and review cadence
Explanation:
Risk scorecards need scenarios, controls, residual risk, owners, metrics, and review cadence. Demo counts, GPU prices alone, or unevidenced green checks are incomplete.

84. A batch nightly job must summarize 100,000 tickets. Which integration approach is most appropriate?

A. One single prompt containing all tickets at once
B. Manual copy-paste into the Claude web UI
C. A controlled batch pipeline with rate limiting, checkpointing, cost budgets, structured outputs, and resumability
D. Open 100,000 interactive chat sessions from a laptop

Correct:
A controlled batch pipeline with rate limiting, checkpointing, cost budgets, structured outputs, and resumability
Explanation:
Large batch work needs pipelines with rate limits, checkpoints, budgets, structured outputs, and resume. Interactive sessions, one mega-prompt, or manual UI work do not scale.

85. Which evaluation approach best detects regressions in a RAG assistant's groundedness?

A. Only measure tokens per second
B. Use answer-groundedness/citation accuracy checks against retrieved evidence plus human spot checks on hard cases
C. Assume groundedness if latency is low
D. Only measure CSS layout shifts

Correct:
Use answer-groundedness/citation accuracy checks against retrieved evidence plus human spot checks on hard cases
Explanation:
Groundedness needs evidence-aligned metrics and human review on hard cases. TPS, CSS, and latency do not measure whether answers stick to sources.

86. What is the primary goal of a discovery workshop for a Claude initiative?

A. To skip stakeholders and start coding agents immediately
B. To select the model brand color
C. To clarify business outcomes, user journeys, constraints, data sources, risks, success metrics, and non-goals before designing the architecture
D. To finalize pixel-perfect Ul before understanding problems

Correct:
To clarify business outcomes, user journeys, constraints, data sources, risks, success metrics, and non-goals before designing the architecture
Explanation:
Discovery aligns outcomes, journeys, constraints, data, risks, metrics, and non-goals before architecture. Ul polish-first, brand color, or skipping stakeholders produce misaligned builds.

87. You need high reliability for a multi-step agent that calls external tools. Which design practice best improves reliability?

A. Increase temperature to 1.0 so the agent invents tool results
B. Define explicit tool contracts, timeouts, retries with backoff, idempotency where needed, and fallback paths when tools fail
C. Remove all logging to reduce overhead
D. Ignore tool errors and always continue generating

Correct:
Define explicit tool contracts, timeouts, retries with backoff, idempotency where needed, and fallback paths when tools fail
Explanation:
Reliable tool-using systems need contracts, timeouts, bounded retries, idempotency for side-effecting calls, and fallbacks. Ignoring errors or inventing tool results is unsafe. Logging is essential for reliability diagnosis.

88. What is a canary release for a Claude application change?

A. Deleting the previous version immediately for all users
B. A marketing announcement without technical rollout controls
C. Shipping to a small percentage of traffic/users first while monitoring quality, errors, latency, and cost before wider rollout
D. Releasing only to canaries in a laboratory

Correct:
Shipping to a small percentage of traffic/users first while monitoring quality, errors, latency, and cost before wider rollout
Explanation:
Canary releases expose a small slice of traffic first with active monitoring before full rollout. They are not animal labs, instant full cutovers, or pure marketing events.

89. How should webhooks or event-driven systems integrate with Claude processing?

A. Process only during manual business hours with no queue
B. Acknowledge quickly, enqueue durable work, process asynchronously with retries/dead-letter queues, and push results via callbacks or status APIs
C. Drop events if the model is busy
D. Block the webhook sender for minutes while the full agent loop completes synchronously

Correct:
Acknowledge quickly, enqueue durable work, process asynchronously with retries/dead-letter queues, and push results via callbacks or status APIs
Explanation:
Event-driven integrations should ack fast, process asynchronously with durable queues and retries/DLQs, then deliver results asynchronously. Long synchronous webhook holds, silent drops, and pure manual windows are unreliable.

90. An integration must fan out the same user question to a search tool, a CRM lookup, and a knowledge base, then synthesize. Which pattern fits?

A. Run tools after sending the final answer to the user
B. Parallel tool calls / parallelization with aggregation and conflict handling in the synthesizer step
C. Ignore CRM and knowledge base to save tokens always
D. Strict single-tool serial only, never parallel

Correct:
Parallel tool calls / parallelization with aggregation and conflict handling in the synthesizer step
Explanation:
Independent retrievals are ideal for parallel tool calls with a later synthesis step that reconciles conflicts. Forced serial-only is slower. Dropping sources by default hurts quality. Running tools after answering prevents grounding.

91. FedRAMP-oriented architecture discussions at a high level emphasize which theme for Claude integrations?

A. Using any consumer app without review for federal data
B. Sharing admin credentials in Slack for speed
C. Authorized cloud environments, continuous monitoring, least privilege, boundary controls, and documented system security plans aligned to the authorization boundary
D. Disabling encryption to simplify assessments

Correct:
Authorized cloud environments, continuous monitoring, least privilege, boundary controls, and documented system security plans aligned to the authorization boundary
Explanation:
FedRAMP-oriented design focuses on authorized environments, continuous monitoring, least privilege, boundaries, and documented security plans. Consumer apps without review, disabled encryption, and shared admin credentials conflict with that posture.

92. Which cost optimization is usually safest for quality-sensitive enterprise answers?

A. Cache stable prefixes, retrieve fewer better chunks, tier models by stage, and measure quality before/after each change
B. Remove all evaluation to save engineer time
C. Disable authentication to reduce overhead
D. Cut the model tier for all stages without measuring quality impact

Correct:
Cache stable prefixes, retrieve fewer better chunks, tier models by stage, and measure quality before/after each change
Explanation:
Safe cost optimization combines caching, better retrieval, model tiering, and measured quality impact. Blind model cuts, removing evals, or disabling auth are not acceptable trade-offs.

93. Which trade-off is most accurate when choosing between a tool-using agent and a pure RAG chatbot for internal knowledge work?

A. Agents always cost less than RAG
B. Agents eliminate the need for evaluation
C. RAG is strong for grounded answers over documents; agents add value when multi-step actions, tools, or dynamic workflows are required-at higher complexity and usually higher cost/latency
D. RAG cannot cite sources while agents always can

Correct:
RAG is strong for grounded answers over documents; agents add value when multi-step actions, tools, or dynamic workflows are required-at higher complexity and usually higher cost/latency
Explanation:
RAG excels at grounded document Q&A. Agents excel when the task needs tools, multi-step actions, or dynamic plans. Agents usually cost more and are harder to evaluate. RAG systems can and should cite sources. Agents increase, not eliminate, the need for evaluation.

94. When is human-in-the-loop evaluation most critical?

A. For high-stakes, ambiguous, or policy-sensitive outputs where automated metrics are insufficient alone
B. Only for deciding button colors
C. Only after a public incident, not before launch
D. Never, because models are always correct

Correct:
For high-stakes, ambiguous, or policy-sensitive outputs where automated metrics are insufficient alone
Explanation:
Human review is essential for high-stakes, ambiguous, or policy-sensitive tasks where automated scores miss nuance. Models are not always correct, and waiting for incidents is too late.

95. Designing a production RAG pipeline, which order of stages is most appropriate?

A. Skip indexing and paste the entire corporate drive into every prompt
B. Ingest/index→ retrieve relevant chunks for the query → ground generation on chunks → cite/validate
C. Train a new foundation model weekly instead of retrieving
D. Generate answer first, then retrieve documents that match the answer

Correct:
Ingest/index→ retrieve relevant chunks for the query → ground generation on
Explanation:
Standard RAG is ingest/index, retrieve for the query, generate grounded on retrieved evidence, then cite/validate. Answering before retrieval is backwards. Weekly foundation training is not RAG. Full-drive prompt dumps do not scale.

96. Which combination best evaluates a tool-using agent?

A. Only number of tools defined in code
B. Only final answer BLEU against a reference poem
C. Whether the demo video looks smooth
D. Task success rate, tool-call correctness, intermediate state checks, cost/latency, and safety violation rate

Correct:
Task success rate, tool-call correctness, intermediate state checks, cost/latency, and safety violation rate
Explanation:
Agent evals need task success, tool correctness, intermediate checks, cost/latency, and safety rates. BLEU on poems, tool counts, or demo polish are insufficient alone.

97. What is the most important reason to version tool schemas in a Claude platform?

A. To make diffs impossible
B. To avoid documenting tools
C. So agents, clients, and servers can evolve safely with compatibility windows and migration plans
D. Because JSON cannot be parsed otherwise

Correct:
So agents, clients, and servers can evolve safely with compatibility windows and migration plans
Explanation:
Versioned tool schemas enable controlled evolution and compatibility. JSON remains parseable without versions; versions improve, not avoid, documentation and safe change.

98. During change management for a new Claude workflow, operations staff resist because they fear job loss and opaque errors. What is the best approach?

A. Remove all human roles immediately
B. Involve operators early, clarify human roles, provide training, expose explainability/override tools, and phase rollout with feedback loops
C. Mock concerns as non-technical
D. Deploy silently without training

Correct:
Involve operators early, clarify human roles, provide training, expose explainability/override tools, and phase rollout with feedback loops
Explanation:
Change management needs early involvement, role clarity, training, overrides/explainability, and phased feedback. Silent deploy, instant job removal messaging, or dismissing concerns create adoption failure.

99. A retailer wants Claude to generate personalized marketing copy. Legal requires brand-safe outputs and marketing wants high volume. Which architecture decision best balances throughput and control?

A. Manual writing only with no Claude involvement
B. Generation pipeline with policy-constrained prompts, safety checks, batch evaluation sampling, and human approval for publish
C. Unrestricted agent with full CRM write access generating and publishing copy automatically
D. A multi-agent system that publishes first and audits later

Correct:
Generation pipeline with policy-constrained prompts, safety checks, batch evaluation sampling, and human approval for publish
Explanation:
High-volume generation is fine when constrained by policy prompts, automated safety checks, sampled evals, and human approval before publish. Full auto-publish with write access is high risk. Manual-only ignores the productivity goal. Publish-first audit-later fails brand-safety requirements.

100. How should prompt and agent configuration changes be managed in a team setting?

A. Edit production prompts live with no review
B. Store prompts/agents as versioned artifacts in git with review, eval CI, and controlled promotion across environments
C. Change prompts via text message to on-call
D. Keep the only copy on one laptop

Correct:

Store prompts/agents as versioned artifacts in git with review, eval CI, and controlled promotion across environments

Explanation:

Prompts and agent configs are production code: version control, review, eval CI, and environment promotion. Live unreviewed edits, laptop-only copies, and SMS changes are operational failures.